



SmartElex High Precision Temperature Sensor TMP117

❖ Introduction –

The **SmartElex TMP117 High Precision Temperature Sensor** is a digital temperature sensing breakout board based on the **Texas Instruments TMP117** sensor. It offers **very high accuracy temperature measurement** with up to **± 0.1 °C accuracy** over a wide temperature range, making it ideal for projects that require precise temperature detection, environmental monitoring, data logging, weather stations, and other precision applications.

❖ Features –

- Uses I2C interface
- Four selectable addresses - 0x48 (default), 0x49, 0x4A, 0x4B
- 16-bit resolution, 0.0078°C
- High accuracy, digital temperature sensor
- ± 0.1 °C (max) from -20 °C to 50 °C
- ± 0.15 °C (max) from -40 °C to 70 °C
- ± 0.2 °C (max) from -40 °C to 100 °C
- ± 0.25 °C (max) from -55 °C to 125 °C
- ± 0.3 °C (max) from -55 °C to 150 °C
- Operating temperature range - -55 °C to $+150$ °C
- Operating voltage range: -1.8 V to 5.5 V
- Typically 3.3 V if using the JST-SH cable
- Low power consumption -3.5μ A (1-Hz conversion cycle)
- 150 nA (shutdown current)
- Programmable operating modes

❖ Specifications –

Specification	Details
Sensor IC	TMP117 (Texas Instruments)
Communication	I ² C (JST-SH)
Operating Voltage	1.8 V – 5.5 V (Arduino friendly)
Accuracy	±0.1 °C (typical from –20 °C to +50 °C)
Resolution	16-bit (~0.0078 °C)
Measurement Range	–55 °C to +150 °C
Interface Address	Default: 0x48 (can change to 0x49, 0x4A, 0x4B)
I ² C Frequency	Standard I ² C speeds (up to 400 kHz typical)

❖ Interfacing with Arduino –

Module Pin	Arduino Uno Pin
VCC	5V
GND	GND
SDA	A4
SCL	A5

❖ Sample Arduino Code –

Ensure the SparkFun TMP117 library is installed.

You can also use functions to read temperature in Fahrenheit or configure alerts if supported by the library.

```
#include <Wire.h>
```

```
#include <SparkFun_TMP117.h> // Library from Arduino Library Manager
```

```
#include <Adafruit_Sensor.h>
```

```
SparkFun_TMP117 tmp117;
```

```
void setup() {
```

```
  Serial.begin(9600);
```

```
  while (!Serial);
```

```
  if (!tmp117.begin()) {
```

```
    Serial.println("TMP117 not detected!");
```

```
    while (1);
```

```
}  
  
Serial.println("TMP117 ready!");  
}  
  
void loop() {  
    float tempC = tmp117.readTempC();  
    Serial.print("Temperature: ");  
    Serial.print(tempC);  
    Serial.println(" °C");  
    delay(1000);  
}
```

❖ Applications –

- Environmental monitoring systems
- Weather stations
- Precision temperature logging
- HVAC control systems
- 3D printer / incubator temperature monitoring
- Scientific instruments and calibration rigs